

Microsoft Windows [Version 10.0.26200.8894]
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C:\Users\vedad>mysql -u root -v
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: NO)

C:\Users\vedad>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> -- STEP 1: CREATE DATABASE
mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql> USE bookflow_db;
Database changed
mysql>
mysql> -- STEP 2: CREATE BOOKS TABLE
mysql> CREATE TABLE Books (
  ->     book_id INT PRIMARY KEY,
  ->     title VARCHAR(100) NOT NULL,
  ->     isbn VARCHAR(20) UNIQUE,
  ->     published_year INT CHECK (published_year < 2027)
  -> );
ERROR 1050 (42S01): Table 'books' already exists
mysql>
mysql> -- STEP 3 & 4: INSERT BOOKS
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
```

```
mysql>
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title       | isbn       | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988          |
| 2       | Clean Code   | 9780132350884 | 2008          |
| 3       | Atomic Habits | 9780735211292 | 2018          |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> -- STEP 5, 6 & 7: CREATE MEMBERS TABLE
mysql> CREATE TABLE Members (
  ->     member_id INT PRIMARY KEY,
  ->     full_name VARCHAR(100),
  ->     email VARCHAR(100) UNIQUE
  -> );
ERROR 1050 (42S01): Table 'members' already exists
mysql>
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name   | email               |
+-----+-----+-----+
| 1         | Anil Kumar  | anil.kumar@example.com |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma  | ravi.verma@example.com |
| 101       | John Smith  | john.smith@email.com   |
| 102       | Emma Wilson | emma.wilson@email.com   |
| 103       | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
6 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> -- STEP 8: CREATE LOANS TABLE
```

```
mysql> CREATE TABLE Loans (
```

```
->     loan_id INT PRIMARY KEY,
```

```
->     member_id INT,
```

```
->     book_id INT,
```

```
->     loan_date DATE,
```

```
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),
```

```
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
```

```
-> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
```

```
mysql> -- STEP 9 & 10: INSERT LOANS
```

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
```

```
-> (1, 101, 1, '2025-01-05'),
```

```
-> (2, 102, 2, '2025-01-08'),
```

```
-> (3, 103, 3, '2025-01-10'),
```

```
-> (4, 101, 2, '2025-02-01'),
```

```
-> (5, 102, 1, '2025-02-05'),
```

```
-> (6, 103, 2, '2025-02-12'),
```

```
-> (7, 101, 3, '2025-03-01'),
```

```
-> (8, 102, 3, '2025-03-07'),
```

```
-> (9, 103, 1, '2025-03-15'),
```

```
-> (10, 101, 1, '2025-04-01');
```

```
Query OK, 10 rows affected (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

```
mysql>
mysql> -- STEP 11: JOIN QUERY
mysql> SELECT m.full_name AS Member_Name,
```

```

->      b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m
-> ON l.member_id = m.member_id
-> INNER JOIN Books b
-> ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Alchemist
Emma Wilson	The Alchemist
Michael Brown	The Alchemist
John Smith	The Alchemist
Emma Wilson	Clean Code
John Smith	Clean Code
Michael Brown	Clean Code
Michael Brown	Atomic Habits
John Smith	Atomic Habits
Emma Wilson	Atomic Habits

```
+-----+
10 rows in set (0.01 sec)
```

```
mysql>
mysql> -- STEP 12: GROUP BY QUERY
mysql> SELECT published_year,
->         COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1988	1
2008	1
2018	1

```
+-----+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> -- STEP 13: CREATE DONATION HISTORY TABLE
mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> -- STEP 14-16: TRANSACTION
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
```

Query OK, 0 rows affected (0.00 sec)

```
mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
    -> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO Donation_History
    -> (donation_id, book_id, donor_name, donation_date)
    -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> -- DISPLAY DONATION HISTORY
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
|          1 |        4 | Raj Kumar  | 2026-08-22    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> -- STEP 17: CREATE INDEX
mysql> CREATE INDEX idx_books_isbn
    -> ON Books(isbn);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> -- STEP 18: FAST SEARCH USING ISBN
mysql> SELECT *
    -> FROM Books
    -> WHERE isbn = '9780451524935';
Empty set (0.00 sec)
```

```
mysql> SELECT *  
-> FROM Books  
-> WHERE isbn = '9780451524935';  
Empty set (0.00 sec)
```

```
mysql>  
mysql> -- FINAL TABLES  
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018
4	Animal Farm	9780451526342	1945

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

6 rows in set (0.00 sec)

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12

```
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1 | 4 | Raj Kumar | 2026-08-22 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql>
mysql> |
```